

INTELLIGENT DEMAND - SHIFTING PLATFORM

RE-FLOW AI

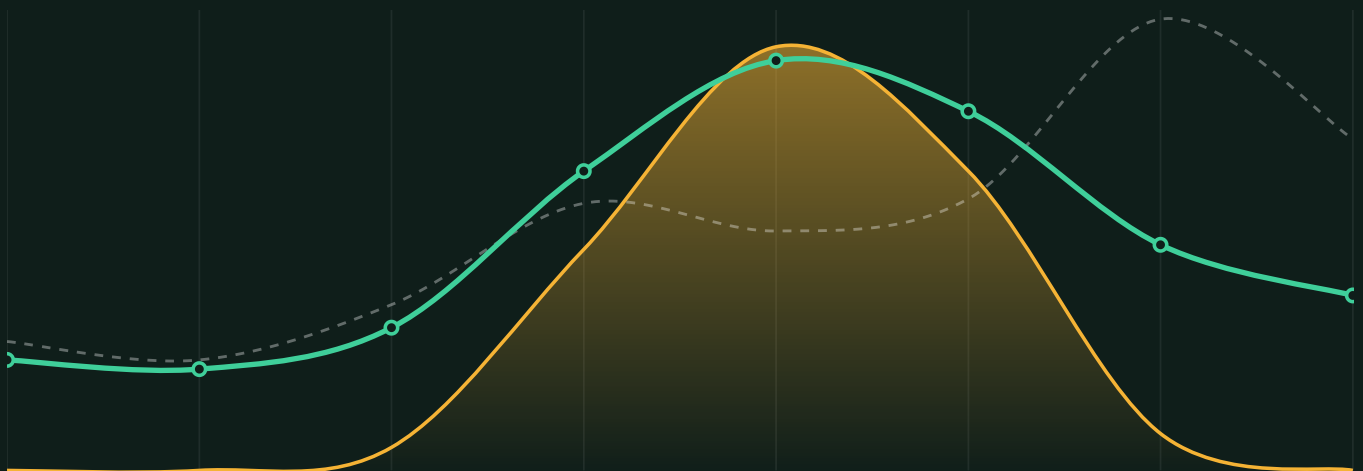
Synchronizing flexible industrial, commercial and EV - charging loads with real-time solar and wind generation.

01 Predict

02 Shift

03 Optimize

04 Save

**WHAT IT DOES**

Turns static energy monitoring into an automated, proactive load-shifting engine.

WHO IT SERVES

Factories, EV fleets, commercial buildings, data centers, BESS and grid operators.

HEADLINE RESULT

Renewable self-consumption up from 42.5% to 78.4% — with a 56.7% cut in carbon intensity.

Consume at a smarter time, not simply less.

01

Most clean-energy tools ask users to cut consumption. RE-FLOW AI solves the **renewable mismatch paradox** instead: the sun peaks at noon, but demand peaks at 6 PM.

By bridging real-time solar and wind generation forecasts with flexible industrial, commercial and EV-charging loads, RE-FLOW AI transforms passive monitoring into an automated decision engine that reschedules energy-intensive work into the hours when clean power is abundant and cheap.

Simulation results below combine industrial process shifting, smart EV fleet charging and battery-storage integration.

“If you need to consume electricity, can you consume it at a smarter time?”

The single question RE-FLOW AI answers — continuously, for every flexible load on the network.

IMPACT AT A GLANCE

Renewable self-consumption

+84.5%

RELATIVE INCREASE

Baseline 42.5%

With RE-FLOW AI **78.4%**

Industrial peak demand

-34.4%

PEAK SHAVING

Baseline 1,250 kW

With RE-FLOW AI **820 kW**

Commercial electricity cost (TOU)

-33.3%

COST REDUCTION

Baseline ₹20.04 /kWh

With RE-FLOW AI **₹13.36 /kWh**

EV fleet renewable charging

+61.5 pts

CLEAN ELECTRONS

Baseline 28.0%

With RE-FLOW AI **89.5%**

Grid carbon intensity

-56.7%

CARBON CUT

Baseline 485 g CO₂/kWhWith RE-FLOW AI **210 g CO₂/kWh**

Renewable curtailment loss

-82.9%

WASTE AVOIDED

Baseline 18.2%

With RE-FLOW AI **3.1%**

Baseline = standard grid operation without load coordination. Cost figures use time-of-use (TOU) commercial tariffs; carbon figures use marginal grid intensity.

Move demand into the solar window.

02

24-hour grid profile

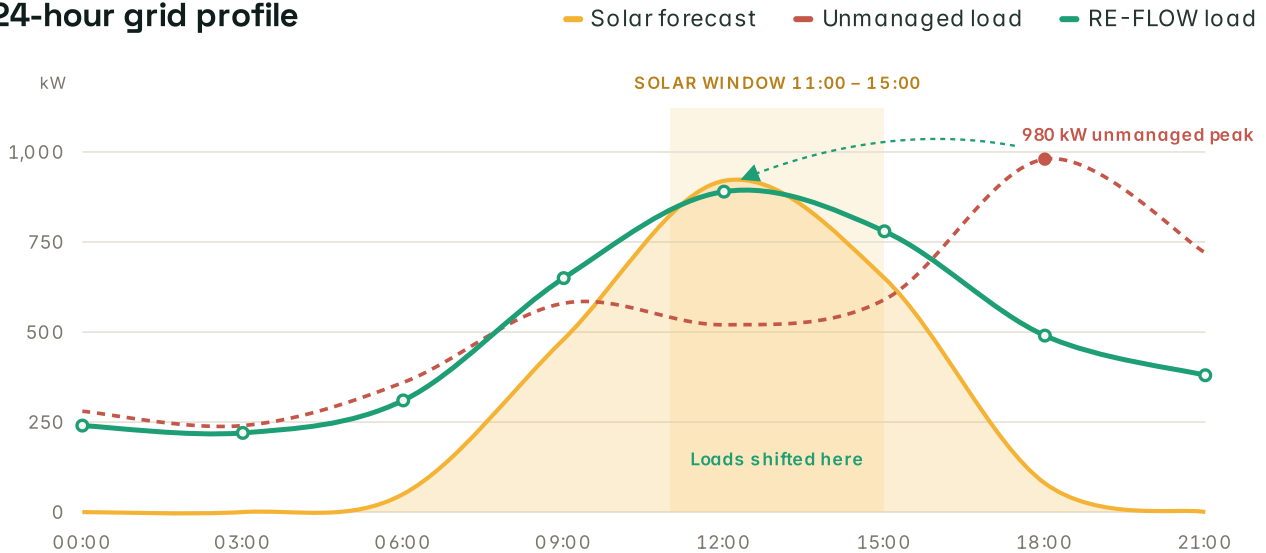


Figure 1. Renewable forecast vs. dynamic load shifting. Flexible loads (EV charging, HVAC pre-cooling, batch machine schedules) are pulled out of the low-solar 6 PM peak and placed into the 11 AM – 3 PM generation window, flattening the evening spike.

980 → 490 kW

Evening peak at 18:00 halved as flexible load is rescheduled.

520 → 890 kW

Midday consumption rises to absorb 920 kW of forecast solar.

Same total energy

Nothing is sacrificed — the work still gets done, just at a cleaner hour.

Renewable energy utilization by sector

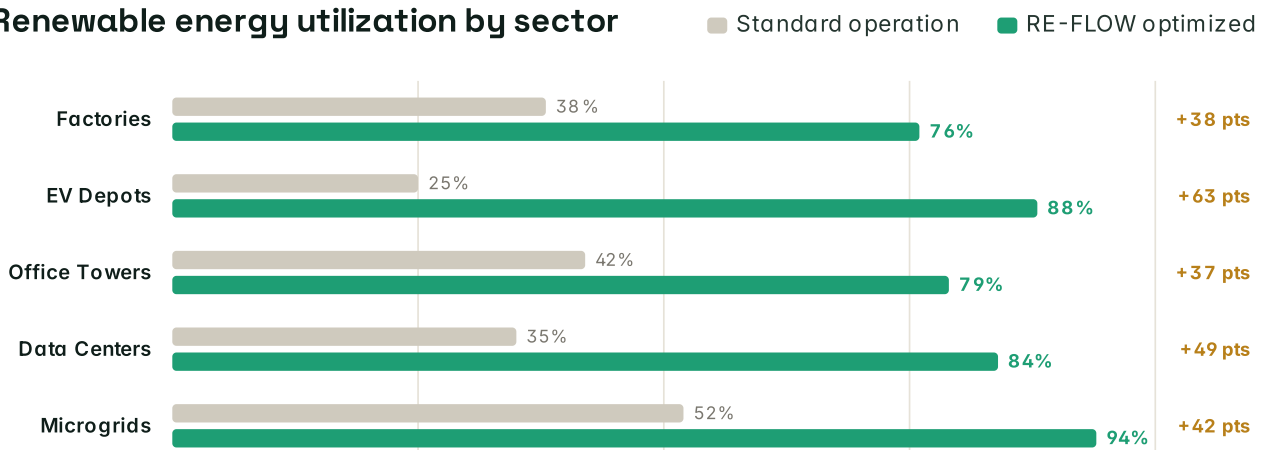


Figure 2. Share of clean energy consumed across key industrial and commercial segments. EV depots see the largest jump because charging is the most time-flexible load of all.

Six places the engine plugs in.

03

01

Industries & manufacturing

High-draw machinery — furnaces, milling, crushing, batch heating — runs on rigid schedules. RE-FLOW AI integrates with existing SCADA / MES systems to identify flexible batch processes that can be shifted safely without compromising delivery deadlines.

EXAMPLE A thermal curing process planned for 6:00 PM (peak tariff, high fossil share) is automatically rescheduled to 1:00 PM, capturing peak solar and shaving demand charges.

02

Smart EV fleets & public charging

EV charging is inherently flexible. Rather than letting dozens of delivery vans charge simultaneously when drivers return at 5:30 PM, RE-FLOW AI staggers and modulates charging rates against solar / wind curves while guaranteeing 100% departure readiness.

EXAMPLE Unmanaged charging at 6:00 PM strains the depot transformer; RE-FLOW AI buffers and schedules fast charging between 11:00 AM – 3:00 PM when rooftop solar produces excess power.

03

Commercial buildings & data centers

HVAC pre-cooling, chilled-water thermal storage, water pumping and heavy batch compute workloads are dynamically shifted to coincide with renewable-rich grid hours — reducing peak building demand by up to 35%.

04

Battery energy storage (BESS)

Instead of blind time-of-day cycling, RE-FLOW AI factors weather forecasts and marginal carbon emissions to charge BESS units strictly when solar / wind generation is highest, discharging precisely during fossil-peaker runtimes.

05

Renewable operators & grid balancers

Moves beyond passive forecasting ("How much solar will we generate tomorrow?") to active supply-demand balancing: "Where and when can flexible loads be activated to absorb this generation without curtailment?"

06

City-scale demand-response ecosystem

Coordinates distributed multi-agent ecosystems across entire cities — factories, EV fleets, commercial towers and smart residential communities — forming a Virtual Power Plant (VPP) that responds dynamically to renewable volatility.

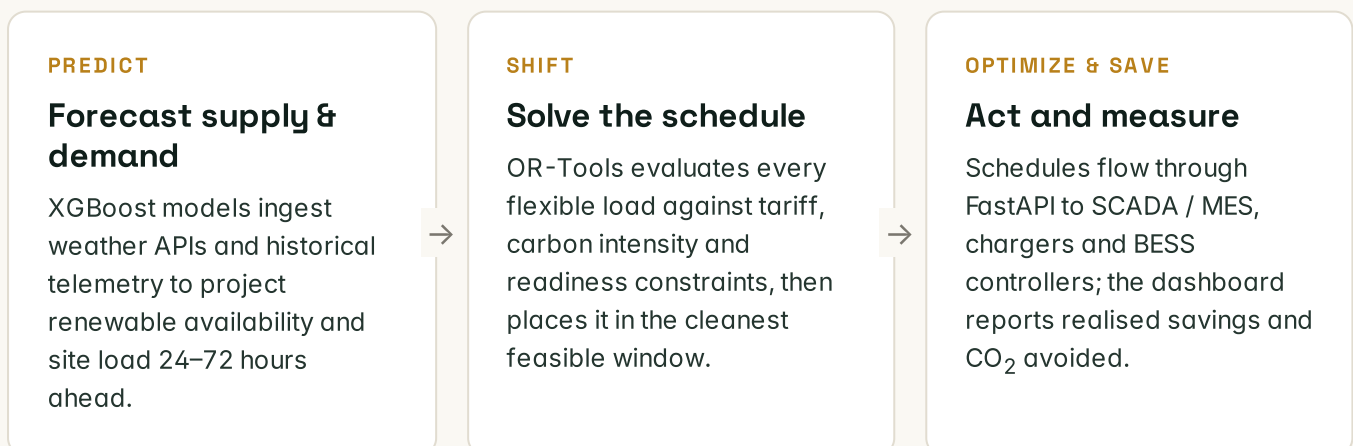
A modular, cloud-native stack with edge connectivity.

04

RE-FLOW AI is engineered as an enterprise-grade platform of independently deployable microservices. Read the stack bottom-up: forecasts and demand predictions feed the optimizer, which writes recommended schedules to the data layer and exposes them through the API and dashboard.

06	Dashboard React · Recharts	Surfaces forecasts, recommended schedules and estimated impact to operators.
05	Backend API FastAPI	Connects the AI models, optimization engine and data layer into one service.
04	Energy data management Python · PostgreSQL	Stores and manages renewable-generation and electricity-demand time series.
03	Load-shift optimization OR-Tools	Finds the best time window to run each flexible electricity load.
02	Demand prediction Python · ML models	Predicts future electricity demand across connected assets.
01	Renewable forecasting Python · XGBoost · Weather API	Predicts renewable-energy availability for the next 24–72 hours.

DECISION PIPELINE



Why RE-FLOW AI wins.

Its core advantage is pragmatism: sustainability without disruption — no operational shutdowns, no sacrifices, just better timing.

STAKEHOLDER	IMMEDIATE OPERATIONAL GAINS	LONG - TERM STRATEGIC IMPACT
End users & fleet managers	- Automated charging scheduling - Zero manual guesswork - Protection against high peak tariffs	- Maximized asset longevity - Guaranteed readiness schedules - Demonstrable ESG compliance
Enterprises & plant operators	30%+ reduction in demand charges - Automated shift recommendations - Seamless ERP / MES sync	- Protection from carbon border adjustments - Lower operational OPEX - Resilient microgrid capability
Grid & energy ecosystem	- Avoided renewable curtailment - Reduced peak transformer stress - Fewer fossil-peaker activations	- Accelerated grid decarbonization - Stabilized frequency & voltage - True 24/7 carbon-free energy (CFE)

FROM MONITORING TO INTELLIGENT ACTION

WHERE MOST PLATFORMS STOP

“Tomorrow, solar generation will be high.”

THE RE-FLOW AI PROMISE

“Tomorrow between 11:00 AM and 3:00 PM, solar will peak at 920 kW. We have scheduled 350 kWh of EV fleet charging, pre-cooled Facility B and shifted 2 industrial batch cycles into this window. Estimated savings: ₹1,42,000 and 1.8 tonnes of CO₂ avoided.”

The decisive final leap.

06

Forecasting tells you what the grid will do. RE-FLOW AI tells the grid what to do with it — turning tomorrow's renewable forecast into today's schedule for every flexible load on the network.

+84.5%

RENEWABLE SELF-USE

-34.4%

PEAK DEMAND

-56.7%

CARBON INTENSITY

-82.9%

CURTAILMENT LOSS



Predict. Shift. Optimize. Save.

An interconnected Virtual Power Plant where factories, fleets and buildings respond to renewable volatility — automatically.